Versioning

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Version schemes

- **Windows**: 1.0, 3.11, 95 (4.0), 98 (4.1), 2000 (5.0), Vista (6.0), 7 (6.1), ... (marketing)

- **Ubuntu**: 22.04 LTS, 22.10, 23.04, 23.10, 24.04 LTS, ... (date based)

- **Numerous software packages:** 0.1.0, 0.1.1, 0.2.0, 1.0.0, 1.1.0, ... (~semver)

Semantic Versioning ("SemVer")

4.2.1

MAJOR *Minor* patch

Given a version number **`MAJOR`**`.`*`Minor`*`.patch`, increment the:

- **`MAJOR`** version when you make <u>incompatible</u> API changes
- *`Minor`* version when you <u>add functionality</u> in a <u>backward compatible</u> manner
- `patch` version when you make backward compatible <u>bug fixes</u>

Additional labels for pre-release and build metadata are available as extensions to the ***`MAJOR`***`.`*`Minor`*`.`*`patch`* *format.*

https://semver.org

Why use SemVer?

- "*This is not a new or revolutionary idea. In fact, you probably do something close to this already. The problem is that “close” isn’t good enough. Without compliance to some sort of formal specification, version numbers are essentially useless for dependency management. By giving a name and clear definition to the above ideas, it becomes easy to* ***communicate your intentions*** *to the users of your software. Once these intentions are clear, flexible (but not too flexible)* ***dependency specifications*** *can finally be made.*"

https://semver.org

Initial development

0.2.1

- **Initial development:** New projects should start with *major version = 0*. Indicates that anything may change; public API is unstable. Usually, the interpretation and risk-level are shifted to the right.

- **Pre-release:** *e.g.* "1.0.0-alpha". Indicates that the version is unstable and might not satisfy the intended compatibility requirements

https://semver.org

Example: `environment.yml`

```yaml
# Conda environment definition
#
#    conda env create -f environment.yml
#    source activate faunusdev
name: faunusdev
dependencies:
    - python=3.8
    - scipy=1
    - ipympl=0.9
    - jinja2=2.10.1
    - numpy>1.17,<1.19.2
    - notebook=5
    - cmake>=3.25
    - ruamel.yaml
```

"pinning"

Task:
Rank according to risk